

Colton Davis

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PURPOSE

I am seeking full-time employment or an internship in Electrical/Software Engineering

EDUCATION

Arizona State University, Tempe Az

Expected May 2026

B.S. Biomedical Device Engineering – Emphasis in Electrical Engineering

- New American University Scholar – President's Award
- Dean's List: Fall 2022 – 2025

Snowflake Highschool, Az

May 2022

- Graduated with Highest Distinction

SKILLS

- | | | |
|---------------------|---------------|----------------------|
| • Circuit Design | • LabVIEW | • Organization |
| • Software Design | • Arduino IDE | • Teamwork |
| • Signal Processing | • Python | • Project Management |
| • Robotics | • Fusion360 | • Problem Solving |
| • 3D Modeling | • LTspice | • Efficiency |

PROJECTS

Mouth-Controlled Computer Mouse

September 2025 – December 2025

- Designed accelerometer circuit and software that inputs cursor control via LabVIEW.
- Developed signal filtering, calibration, and a sensitivity interface for further user customization and use.

August 2025 – September 2025

Pulse Oximeter

- Designed and created a prototype to monitor blood oxygen using a stretch sensor and myDAQ system.
- Integrated LabVIEW controls to further data acquisition, exported the data in excel for further statistical analysis.

January 2025 – May 2025

Tesla Valve System Mathematical and 3D Models

- Designed and developed mathematical models for a tesla valve system aimed at hospital catheter systems improvement.
- Conducted research into FDA regulations, regulatory pathways, and laws to ensure federal compliance.

January 2023 – May 2023

Arduino Uno Automated Pill Dispenser

- Designed and created a robotic drug dispensing system based on sensor data and controlled via Arduino circuit and code.
- Implemented software customization and actuation of servos dependent on a given input to prove concept.

VOLUNTEERING

Engineering Projects in Community Service

August 2023 – June 2025

- Led and collaborated with engineers to apply engineering principles to solve real-world problems.
- Designed and delivered a smart composter to Clark Park Community Garden utilizing ESP-32 circuit with wireless and emailing capabilities.

RELEVANT COURSES

- | | |
|--|---|
| • Engineering Capstone I & II (BME 490, 491) | • Biomedical Instrumentation (BME 413, BME 423) |
| • Circuits (EEE 202) | • Numerical Methods (BME 301) |
| • Microcomputer Applications (BME 370) | • Thermodynamics (BME 340) |
| • Signals & Systems for Bioengineers (BME 350) | • Introduction to Python (CSE 194) |